

week-1

Day 1

- Introduction to the module and getting to know each other as a group

Day 2

- Started with p5live
- First basic drawing code

```
function setup() {  
  createCanvas(windowWidth, windowHeight)  
  background(0,0,255)  
  
}  
  
function draw() {  
  //very basic drawing program  
  ellipse(mouseX,mouseY,20)  
}
```


- Basic squares

```
function setup() {  
  createCanvas(windowWidth, windowHeight)  
  background(0)  
}  
  
function draw() {  
  fill(255,0,100)  
  stroke(255)  
  square(300,200,100)  
  strokeWeight(10)  
  
  fill(20,0,100)  
  stroke(255)  
  square(400,200,200)  
  strokeWeight(10)  
  
  fill(100,0,10)  
  square(600,200,300,50)
```

```
}
```


- basic loop

```
function setup() {
  createCanvas(windowWidth, windowHeight)
  background("#EDE8D0")
  rectMode(CENTER)
}

let dim = 200
let reduction = 20
let posX = 400
let posY = 400

function draw() {
  noFill()
  strokeWeight(1)
  stroke(0)
  square(posX, posY, dim)
  square(posX, posY, dim-(reduction*1))
  square(posX, posY, dim-(reduction*2))
  square(posX, posY, dim-(reduction*3))
  square(posX, posY, dim-(reduction*4))
  square(posX, posY, dim-(reduction*5))

  //loop
  for(let i = 0; i < 10; i++){
    square(posX + -100, posY, dim - (reduction * i))
  }

  for(let i = 0; i < 10; i++){
    square(posX, posY-100, dim - (reduction * i))
  }
  for(let i = 0; i < 10; i++){
    square(posX + 100, posY, dim - (reduction * i))
  }
  for(let i = 0; i < 10; i++){
    square(posX, posY+100, dim - (reduction * i))
  }
}
```


- square animation

```
function draw() {
  background("#ff00ff")
  noFill()
  strokeWeight(2)
  stroke(0)
  dimX = 500 + (sin(frameCount*4)* 20)
  num = 17
  reduction = dimX / num
  //loop
  for(let i = 0; i < num; i++){
    let offsetX = (random(15))
    let offsetY = (random(15))
    stroke(0,0,255)
    if (i < 7){
      stroke("#00ebf4")
    }
    square(
      posX + offsetX,
      posY + offsetY,
      (dimX) - (reduction * i)
    )

  }

}
```


•

Day 3

- moving square loop

```
// {"P5LIVE":{"name":"basic-molnar-animation","mod":1777388069280}}
// declare initiale variables for drawing a series of concentric
squares
let dimX = 453
let dimY = dimX
let num = 17
let reduction = dimX / num
let posX = 0
let posY = 0
```

```

// initialize the sketch
function setup() {
  createCanvas(windowWidth, windowHeight)
  background(255)
  // the following functions are very important for
  // setting the squares origin point
  rectMode(CENTER)
  // and the angles to be calculated in the 0 - 360 mode
  // and not in radians 0 - 2*PI
  angleMode(DEGREES)
  // set initial posX and posY to half the canvas
  // so that the square sits in the center of the canvas
  posX = width / 2
  posY = height / 2
}

function draw() {
  background(255)
  noFill()
  strokeWeight(3)
  stroke(0)

  // for loop
  for(let i = 0; i < num; i++) {
    let offsetX = random(7)
    let offsetY = random(7)
    stroke(255, 0, 0)
    strokeWeight(3)
    if(i == 0) {
      stroke(0, 0, 255)
      strokeWeight(1)
    }

    if(i == 4) {
      stroke(0, 0, 255)
      strokeWeight(1)
    }
    square(
      posX + offsetX - 100,
      posY + offsetY,
      (dimX) - (reduction * i)
    )
  }

  dimX = 300 + sin(frameCount * 4) * 10
  num = 10
  reduction = dimX / num
}

```

```

    tmcs(300, 750, 300, 5, 10)
    tmcs(950, 750, 300, 10, 5)

}

// this function draws squares at position
// X and Y and they move randomly a bit
function tmcs(x, y, dim, num, speed) {
    let dimension = dim + sin(frameCount * speed) * 10
    let reduction = dimension / num

    for(let i = 0; i < num; i++) {
        let offsetX = random(7)
        let offsetY = random(7)
        stroke(255, 0, 0)
        strokeWeight(3)
        square(
            x + offsetX,
            y + offsetY - 300,
            (dimension) - (reduction * i)
        )
    }
}

```


- square pattern

```

let dimX = 453
let dimY = dimX
let num = 17
let reduction = dimX / num
let posX = 0
let posY = 0

function setup() {
    createCanvas(windowWidth, windowHeight)
    background("#ff00ff")
    rectMode(CENTER)
    angleMode(DEGREES)
    posX = width / 2
    posY = height / 2
}

function draw() {
    background("#ff00ff")
    noFill()
    strokeWeight(3)
    stroke(0)
}

```

```

let numX = 10

for(let i = 0; i < numX; i++) {
  let dimension = width / numX
  let posX = dimension / 2 + (i * dimension)
  for (let j = 0; j < numX; j++){
    let posY = dimension / 2 + (j * dimension)
    tmcS(posX, posY, 4, dimension, 10)
  }
}

// this function draws squares at position
// x and y, and they move randomly
// a tiny bit
function tmcS(x, y, speed, dim, num) {

  let dimension = dim + sin(frameCount * speed) * 10
  // num = 10
  let reduction = dimension / num

  for(let i = 0; i < num; i++) {
    let offsetX = random(7)
    let offsetY = random(7)
    stroke(255)
    strokeWeight(3)
    square(
      x + offsetX,
      y + offsetY,
      (dimension) - (reduction * i)
    )
  }
}

```


- moving box

```

// {"P5LIVE":{"name":"moving box","mod":1777472826881}}

let posX = 0
let posY = 0
let boxDim = 100

function setup() {
  createCanvas(windowWidth, windowHeight, WEBGL)
}

```

```

function draw() {
  background(255)
  orbitControl()
  fill(255)
  posX++
  posY = posY - 10
  if (posX > (width/ 2) + (boxDim / 2)){
    //posX = 0
    posX = - (width/2)
  }

  if (posY < - (height / 2 )){
    posY = 0
  }

  push()
  translate(posX,posY,0)
  stroke(0)
  box(boxDim)
  pop()

  stroke (255,0,0)
  strokeWeight(5)
  push()
  line(0,0,0, width, 0,0)
  stroke (0,255,0)
  line(0,0,0, 0, -height,0)
  stroke (0,0,255)
  line(0,0,0, 0, 0,1000)
  pop()

}

```


- dvd box animation in a 3d space

```

// {"P5LIVE":{"name":"dvd box animation","mod":1777637555186}}

let x;
let y;
let z;
let xspeed;
let yspeed;
let zspeed;
let depth = 2000;
let dvd;
let r, g, b;

```

```

function setup() {
  createCanvas(windowWidth, windowHeight, WEBGL);
  //initianize variables for the positions of
  //the cube at the beginning
  x = random(width);
  y = random(height);
  z = random(-depth, 0)
  //initialize speed at wich the cube moves
  xspeed = 4;
  yspeed = 4;
  zspeed = 4;

  pickcolor();
}

function pickcolor() {
  r = random(255);
  g = random(255);
  b = random(255);
}

function draw() {
  translate (-width/2, -height/2)
  background(0);
  noStroke(255);
  fill(255);
  fill(r, g, b);

  push()
  //rect(x, y, 80, 60);
  translate (x,y,z)
  stroke(0)
  box(200)
  pop()
  //image(dvd, x, y, 80, 60);

  x = x + xspeed;
  y = y + yspeed;
  z = z + zspeed;

  if (z >= 0) {
    zspeed = -zspeed;
    z = 0;
    pickcolor();
  } else if (z <= -depth) {
    zspeed = -zspeed;
    z = -depth;
  }
}

```

```

    pickcolor();
}

if (x >= width) {
    xspeed = -xspeed;
    x = width;
    pickcolor();
} else if (x <= 0) {
    xspeed = -xspeed;
    x = 0;
    pickcolor();
}

if (y >= height) {
    yspeed = -yspeed;
    y = height;
    pickcolor();
} else if (y <= 0) {
    yspeed = -yspeed;
    y = 0;
    pickcolor();
}
}

```


Day 4

What we did

- Yann showed us how to document with markdown
- Downloaded vcv Rack
- Got started with vcv Rack
- Created a siren like sound

- Created a basic synthesizer controlled with the keyboard

Day 5

- Meeting with Yann: talked about the mid term presentation
- Catching up on the material from yesterday afternoon's class

